

Machine learning for data analytics

Assignment 5: Generative Models

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Each assignment should be submitted as a Jupyter notebook with complete code, visualizations, and written analysis.

Problem 5.1: Variational Autoencoder (35 points)

Implement VAE from scratch in PyTorch:

- Train on Fashion-MNIST dataset.
- Implement reparameterization trick correctly.
- Visualize latent space in 2D using t-SNE.
- Generate new samples by sampling from latent distribution.
- Perform latent space interpolation between pairs of images.

Problem 5.2: GAN Implementation (35 points)

Build a DCGAN for image generation:

- Train on CIFAR-10 dataset.
- Implement proper generator and discriminator architectures.
- Monitor training stability using loss curves and Fréchet Inception Distance.
- Generate grid of synthetic images throughout training.
- Discuss challenges encountered and solutions applied.

Problem 5.3: Anomaly Detection with VAE (30 points)

Use VAE for unsupervised anomaly detection:

- Train VAE on normal samples from credit card fraud dataset.
- Use reconstruction error as anomaly score.
- Compare with traditional anomaly detection methods (Isolation Forest, One-Class SVM).
- Evaluate using precision-recall curves and ROC-AUC.